Bubble sort

Bubble sort compares neighboring items and swaps an adjacent pair when they are out of order. A full ascending-order pass moves the largest remaining item to the end of the unsorted region. A pass with no swaps permits an early stop.

Selection sort

Selection sort scans the unsorted region to find its smallest item, then swaps that item into the first unsorted position. It expands the sorted region by one position on each pass. The usual swapping version does not preserve equal items' original order.

Insertion sort

Insertion sort maintains a sorted prefix. For the next item, shift larger prefix items one position right and insert the item into the gap. Shifting only strictly larger items preserves the order of equal keys.

Merge sort

Merge sort divides the sequence into halves, recursively sorts both halves, and merges the two sorted sequences. Standard array merge sort takes O(n log n) time and O(n) auxiliary storage. Taking from the left half first when keys are equal makes the merge stable.

Quicksort partition

Quicksort chooses a pivot and partitions the remaining items around that pivot before recursively sorting the partitions. Balanced splits give O(n log n) time; repeatedly extreme splits can give O(n squared) time. It does not use merge sort's combine-two-sorted-halves step.